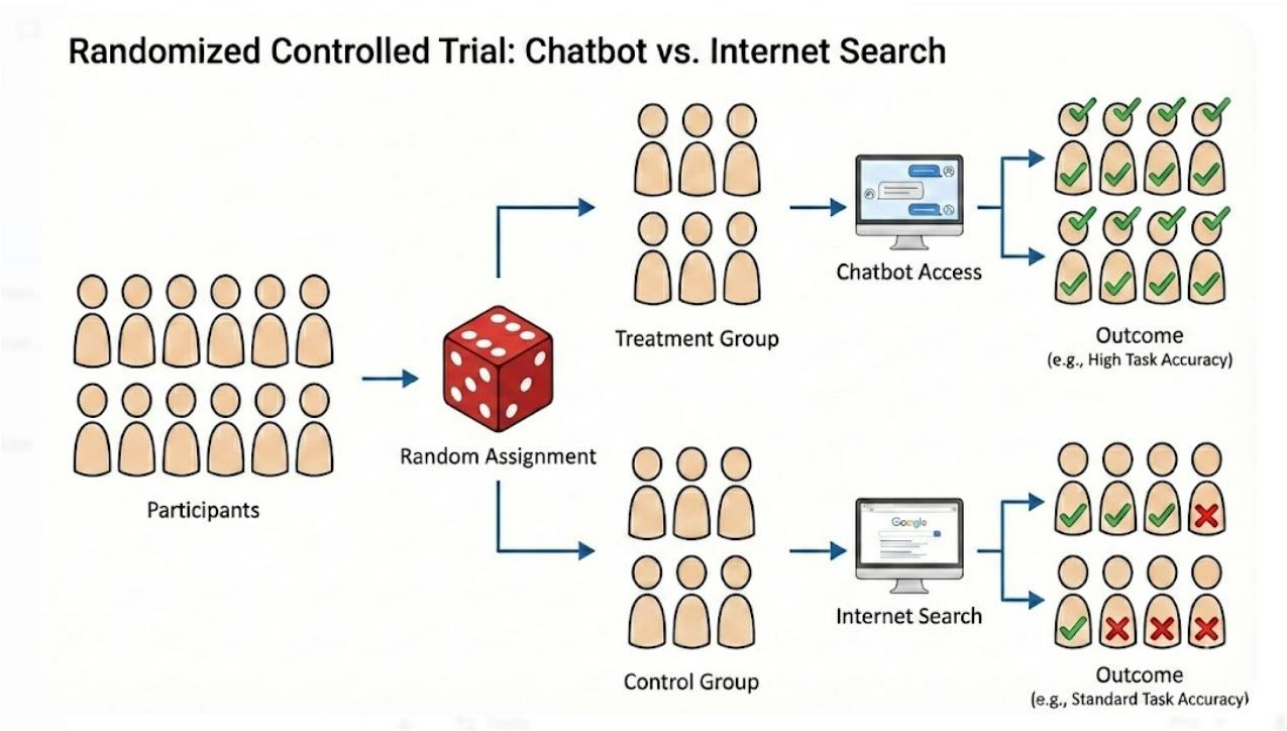
Randomized Controlled Trial (RCT): What are the key characteristics? by Carsten Bergenholtz (ChatGPT-5.2 Thinking and Gemini 3.0 Pro used for editing), 10.01.26

If we want to know whether an intervention or treatment works (e.g., whether giving people access to a chatbot improves performance), we cannot simply compare users with non-users. Maybe users are different from non-users in other ways (e.g. in terms of ability, motivation, study habits, etc.). Furthermore, if we just give everyone access to a chatbot to see if they change performance over time, we can't know if other factors also changed in the meantime (see more details in Isager 2023). The internal validity of such comparisons would be weak. **Internal validity** means how confident we can be that a particular research study establishes something that is true.

An RCT (randomized controlled trial) is a research design that enables one to be more confident about estimating the causal effects of such interventions. The key elements of an RCT are visualized in the following image. Properly understanding all elements is important.



Built by Gemini 3.0 Pro, based on https://www.simplypsychology.org/randomized-controlled-trial.html

The aim of this text is to explain the core elements of an RCT, and why it constitutes a particular research design; some categorize it as the gold standard, for establishing causality. First, I explain a) how a RCT allows us to compare an outcome with the counterfactual, then I present b) the key elements of an experiment (randomization, treatment and control), c) how to assess the validity of RCTs and d) some ethical considerations. It is important to get an understanding of a-d. Note that we in this text focus on people as being part of experiments, but experiments can also involve teams, firms, regions, webpages and other kinds of units of analysis.

Causality: RCT allows a comparison of an outcome with the counterfactual

In research, we are often interested in whether an independent variable (e.g. getting access to chatbots while doing an exam) impacts a dependent variable (e.g. performance in an exam). In a RCT, we 'manipulate' the **independent variable (IV)** to determine if it impacts the **dependent variable (DV)**. This manipulation is the treatment (or intervention). We then compare a treatment group (or multiple treatment groups) with a control group. In this way the control group constitutes a reasonable counterfactual: the students in the above example face identical situations, except for the treatment – hence we can isolate the impact of the treatment, by looking at what happens when students are in the control group.

Examples of Treatments and Control Groups

1. *Aspirin on headache*: One group receives aspirin while the other group receives a placebo. By comparing the two, we can determine if aspirin has a significant effect on headache relief. Both groups live their normal regular life, so on average they should all experience the same things – except for the treatment, which is the only difference.
2. *Size of incentives and performance*: While performing a number of tasks, one group receives a small (monetary) incentive, while another group receives a larger (monetary) incentive. Since they performed the same tasks in the same environment, the difference in their performance levels should reveal the effect of incentive size on performance (this one is used as an example in Bergenholtz 2026b, curriculum).

However, not any comparison is valid. We cannot, for example, simply compare individuals who **themselves** decided to use chatbots to those who did not — this could introduce **selection bias** and other **confounding** factors. For example, maybe those that select into using chatbots are different from those that do not. They could be smarter, dumber, lazier – confounding factors in the sense that they could disrupt the opportunity to interpret an effect on the dependent variable to stem from the independent variable. Instead, an RCT ensures that the treatment is the only difference between the groups, by enforcing a **random assignment.**

Randomization

In essence, by **randomizing** people into **treatment** and **control** groups, we ensure that the two groups are *as similar as possible* - and the only difference is the treatment. Imagine a lab-experiment where we randomize people into two groups; they all have to do the same tasks in the same environment, just with one group having access to a chatbot, and the other not having access to a chatbot. In this way, the randomization aims to ensure that the control group constitutes a reasonable counterfactual – what would have happened if the chatbot wasn't accessible.

Random assignment is designed to make the groups comparable, but "comparable" is a statistical idea rather than a guarantee in every single study. In small samples, chance imbalances are common; in larger samples, randomization tends to produce groups that are much more similar (see this https://carstenbergenholtz.github.io/Phil-of-Science-tools/randomization_sample_size.html for an interactive example showing this point). A coin flip illustrates the logic. If you flip a coin 20

times, you might get 13 tails and 7 heads, a big difference. But if you flip it 150 times, the result will usually be closer to a 50/50 split—say, 77 heads and 73 tails. Likewise, when we randomize a reasonably large number of participants, we can be more confident that the treatment and control groups are balanced on characteristics like age, gender, and prior ability—so the main systematic difference between them is the treatment.

The following table illustrates the point; note the sample size is 100:

	Group	Age	Experience (Years)	Motivation Score	Female	Male
1	Control	25.0	3.8	7.4	29	21
2	Treatment	24.1	3.7	7.5	26	24

(Table and randomization generated by ChatGPT-4).

The distribution is not perfect; there are some differences, but they are small and we can assume that they do not matter. In research studies we often make such a numerical comparison of key characteristics of participants in an experiment, to verify randomization has worked properly.

In Isager's (2023) terms, random assignment guards against both **selection bias** and **confounding**, supporting clear causal direction and high internal validity. Because participants do not self-select into the conditions (e.g., "chatbot fans" ending up disproportionately in the chatbot group), selection is less likely to create systematic group differences. Put differently, selection bias can generate imbalances between groups, and those imbalances can become confounding: pre-existing differences in ability, motivation, or prior experience could otherwise distort the treatment–control comparison.

Treatment allocation (allows comparison to control group)
Once participants are randomized into their respective groups, the treatment (the variable being tested) is applied to the treatment group, while the control group does not receive the treatment. In some cases, the control group might receive a placebo, depending on the study design. Note that a study can involve several treatment conditions – as long as there is always only one difference between any condition. For example, one could give one group access to ChatGPT, another to Google Gemini, a third the internet, and the fourth no technological assistance.

Consider a chatbot example in further detail. Participants in the treatment group are given access to the chatbot, while students in the control group continue studying without it. Everything else, such as the study environment and materials, remains the same for both groups. This step ensures that the treatment is the only variable that differs between the groups, making it easier to determine whether the treatment (chatbot access) has a causal impact on the outcome. In essence, the control group is the counterfactual – what would have happened, on average, without the treatment. If the randomization process had not worked, and the people in the groups were not basically identical, then we would risk that it is not the treatment that leads to different outcomes, but (confounding) differences between people in the respective groups.
It is important that the treatment and control groups remain meaningfully different, which requires limiting interaction between them about the intervention. If people in the treatment group share the tool, outputs, or strategies with people in the control group—or if we cannot rule out such

communication—the control group is partly "treated" as well. This contamination blurs the contrast between groups and makes the estimated treatment effect less clear-cut (and typically smaller).

Measuring outcome and analysis of RCT

To compare the treatment and control conditions, you need a clearly defined outcome (e.g., performance on a task, measured after completion). You then analyze the outcome data quantitatively by estimating the difference between the groups—for example, a difference in average scores. The result is typically an **average treatment effect** across participants. So, if the chatbot group performs better on average than the control group, we conclude that chatbot access improved performance on average. This does not imply that every treated individual performs better than their own counterfactual; it means that, on average, the treatment has a positive impact. See following plot for illustration.

Static figure from one simulation (fixed seed). Dots are individuals. Lines are fitted (OLS) within each group. Key point: overlap remains even when the treatment has a positive average effect.

Produced by Bergenholtz & ChatGPT 5.2 Thinking

To summarize

An RCT randomly assigns comparable units to a treatment or control condition. The treatment group receives the intervention; the control group does not (or receives a placebo). By keeping everything else the same and comparing outcomes, we estimate the average causal effect and approximate the counterfactual (what would have happened without the treatment)

Potential weaknesses: When should you be skeptical of an RCT result?

Even in a randomized controlled trial, there are reasons not to fully trust the estimated effect:

- Randomization may not have worked well in practice (especially with small samples): the groups can still differ by chance on important characteristics.

- Noncompliance and attrition: people may not follow their assigned condition, or they may drop out differently across groups, which can bias the comparison.
- Contamination / spillovers: the control group may partly receive the treatment indirectly (e.g., colleagues share the tool or advice), reducing the meaningful difference between groups.
- "Not only one difference": apart from the treatment, other factors may differ across groups (different instructions, timing, manager attention, resources), undermining the causal interpretation.

Bergenholtz 2026b presents a more detailed illustration of what an actual RCT looks like in practice, including some insights on typical challenges as well as an illustration of how to assess the external validity, cf. next section.

External validity criteria
The prior sections have explained what a RCT is and why it has high internal validity (that is, why we can be confident that we have identified a clear causal effect in the particular sample).
To ensure findings are applicable beyond the specific context of the study, the external validity should be considered. **External validity** involves assessing if results can be transferred to or generalized to other situations (e.g. other empirical settings).
Four core criteria must be put in play to evaluate external validity effectively:

1. **Interaction of selection and treatment:** What group of people were part of the experiment, and can the results apply to different groups of people?
2. **Interaction of setting and treatment:** In what setting did the experiment take place. E.g., what task should the participants complete in the experiment?
3. **Interaction of history and treatments:** Society changes. Would the results hold true at a different time? E.g., might results from 1960 hold true for 2025?
4. **Hawthorne effect:** Might the results be shaped by people knowing they were part of an experiment? Recall insights from Organizational Behavior, and that people might change behavior if they know they are being observed.

(list of criteria based on Bell et al. 2022, p. 54-55).

Keeping these four elements in mind is central when evaluating the external validity of an RCT—**so if you are asked to assess external validity, use them as your checklist**. One related concept is **ecological validity**. Whereas external validity concerns whether findings generalize to other people, settings, and times, ecological validity is specifically about how **naturalistic** the study context is: does the experiment resemble the real-world situation we care about?
This distinction becomes concrete when we compare **lab and field experiments**. **Lab experiments are conducted in controlled, often artificial environments** (e.g., the COBE Lab at BSS). This control typically strengthens internal validity because it is easier to isolate the treatment effect and keep everything else constant. The downside is that the task and setting may be less realistic—for example, giving participants chatbot access for a one-hour task in a lab-experiment may not resemble how people use tools in actual jobs in a real organization over weeks or months. The (more) constrained lab-setup can thus reduce ecological validity, compared to the more realistic field-setup.
Field experiments are conducted in natural settings (e.g., in an actual organization). Because participants do real work in real contexts, **field experiments often have higher ecological validity**, and the outcomes may be more realistic and potentially more generalizable. However, researchers

have **less control**: it can be difficult to prevent spillovers (e.g., treatment and control participants might talk to each other over lunch about their experiences), and it is harder to hold all other factors constant. This can weaken internal validity relative to a lab setting – note that a field-experiment is expected to have a far stronger internal validity than a cross-sectional study.

In short, lab experiments prioritize control (often higher internal validity) while field experiments prioritize realism (often higher ecological validity, and sometimes stronger external validity). For example, you can study the effect of incentives on performance using artificial tasks in a lab or through changes in incentives in a real workplace. The lab makes it easier to ensure the only difference is the treatment; the field better reflects real-world complexity, but makes it harder to rule out spillovers and other influences.

Quasi-Experiments

Moving beyond the pure experiment, we can consider research designs that are almost as good as an actual experiment; quasi-experiments. These resemble RCTs but lack full internal validity, often due to the absence of randomization. They are "studies that have certain characteristics of experimental designs but do not fulfil all the internal validity requirements" (Bell et al. 2022: p. 57). See Bergenholtz 2026c (curriculum), for a more in-depth presentation of what quasi-experiments are (not), including a few examples of quasi-experiments.

Limitations and ethical concerns of RCTs

Practical Challenges: Conducting RCTs in business settings presents several **logistical** and **methodological challenges** that can limit their effectiveness. One common issue is the risk of contamination between treatment and control groups. In a workplace, employees often communicate across groups, sharing information or behaviors that can affect the experiment's outcomes. For example, if one group is given access to a new software tool, they might discuss its benefits with colleagues in the control group, thereby reducing the contrast between the two groups and weakening the study's validity.

Moreover, business settings are often **complex** and shaped by many external influences—economic conditions, company culture, and market dynamics, for example. Because so many factors change, it becomes more difficult to isolate the effect of the treatment and can weaken internal validity (we return to this issue in the lecture on complexity). In addition, many organizational variables are difficult to manipulate experimentally. Culture or leadership style, for instance, typically changes slowly and cannot be randomized or tightly controlled over short periods. Finally, organizations may resist the disruption that RCTs can create, because running an experiment often requires changes to workflows, staffing, or resource allocation in an already busy and evolving environment.

These challenges highlight that while RCTs are powerful for establishing causality, they are not always suitable for every business research question. The complex and dynamic nature of business environments often means that we need to supplement RCTs with other research methods, such as longitudinal studies, qualitative approaches, or observational data, to capture a fuller picture of the factors at play.

Ethics: While RCTs are a powerful tool for establishing causality, they can raise significant **ethical concerns** in business settings. One key issue is the potential harm or disadvantage to participants in the control group, particularly in organizational experiments. For instance, testing new employee policies—such as a more flexible work schedule—might benefit the treatment group while leaving

the control group at a disadvantage. Not being in the treatment (or control) group could lead to job dissatisfaction. Moreover, withholding positive interventions, like access to leadership development programs, could be seen as unethical if the organization already suspects these programs to be helpful. In such cases, researchers must carefully consider the ethical implications of not letting everyone get access to potential benefits.

Additionally, employees in a business setting are often not in the same position as voluntary participants in psychological or medical trials. They may feel forced to participate in the experiment due to workplace pressures or fear of reprisal, which raises concerns about informed consent. Ensuring that participation is truly voluntary and that participants fully understand the study's risks and benefits is essential. Researchers need to be transparent about the purpose of the RCT and guarantee that no employee will face negative consequences for their role in the study, whether they are in the treatment or control group.

References

Bell, Emma, Alan Bryman, and Bill Harley (2022). *Business research methods*. Oxford University Press, 2022

Isager (2023): From curriculum – see https://pedermisager.org/blog/why_does_correlation_not_equal_causation/

If you plan to cite this document, e.g. in a bachelor project, use one of these references:

Bergenholtz, C. (2026a). Randomized Controlled Trial (RCT): What are the key characteristics? Course material, Aarhus University Brightspace.

Bergenholtz, C. (2026a). *Randomized Controlled Trial (RCT): What are the key characteristics?* [Course material]. GitHub. https://github.com/CarstenBergenholtz/Phil-of-science-docs/blob/main/Bergenholtz_2026a_RCTs.pdf